Using this equation we obtain for $1 \leqslant t \leqslant T$:

$$\begin{aligned}
\kappa(0, t-1) &= \mathrm{E}_{s_t, a_t, R_{t+1}} \left[R_{t+1} \mid s_{t-1}, a_{t-1} \right] \\
&= \mathrm{E}_{s_t, a_t} \left[\tilde{q}^{\pi}(s_t, a_t) \; - \; \tilde{q}^{\pi}(s_{t-1}, a_{t-1}) \mid s_{t-1}, a_{t-1} \right] \\
&= \sum_{s_t, a_t} p(s_t, a_t \mid s_{t-1}, a_{t-1}) \; \left(\tilde{q}^{\pi}(s_t, a_t) \; - \; \tilde{q}^{\pi}(s_{t-1}, a_{t-1}) \right) \\
&= \tilde{q}^{\pi}(s_{t-1}, a_{t-1}) \; - \; \sum_{s_t, a_t} p(s_t, a_t \mid s_{t-1}, a_{t-1}) \, \tilde{q}^{\pi}(s_{t-1}, a_{t-1}) \\
&= \tilde{q}^{\pi}(s_{t-1}, a_{t-1}) \; - \; \tilde{q}^{\pi}(s_{t-1}, a_{t-1}) \; = \; 0 \, .
\end{aligned} \tag{A77}$$

Next, we consider the expectation of $\sum_{\tau=0}^{m} R_{t+1+\tau}$ for $1 \leqslant t \leqslant T$ and $1 \leqslant m \leqslant T - t$ (for $m > 0$)

$$\begin{aligned}
\kappa(m, t-1) &= \mathrm{E}_{\pi} \left[\sum_{\tau=0}^{m} R_{t+1+\tau} \mid s_{t-1}, a_{t-1} \right] \\
&= \mathrm{E}_{\pi} \left[\sum_{\tau=0}^{m} \left(\tilde{q}^{\pi}(s_{\tau+t}, a_{\tau+t}) \; - \; \tilde{q}^{\pi}(s_{\tau+t-1}, a_{\tau+t-1}) \right) \mid s_{t-1}, a_{t-1} \right] \\
&= \mathrm{E}_{\pi} \left[\tilde{q}^{\pi}(s_{t+m}, a_{t+m}) \; - \; \tilde{q}^{\pi}(s_{t-1}, a_{t-1}) \mid s_{t-1}, a_{t-1} \right] \\
&= \mathrm{E}_{\pi} \left[\mathrm{E}_{\pi} \left[\sum_{\tau=t+m}^{T} \tilde{R}_{\tau+1} \mid s_{t+m}, a_{t+m} \right] \mid s_{t-1}, a_{t-1} \right] \\
&- \mathrm{E}_{\pi} \left[\mathrm{E}_{\pi} \left[\sum_{\tau=t-1}^{T} \tilde{R}_{\tau+1} \mid s_{t-1}, a_{t-1} \right] \mid s_{t-1}, a_{t-1} \right] \\
&= \mathrm{E}_{\pi} \left[\tilde{R}_{T+1} \mid s_{t-1}, a_{t-1} \right] \; - \; \mathrm{E}_{\pi} \left[\tilde{R}_{T+1} \mid s_{t-1}, a_{t-1} \right] \\
&= 0 \, .
\end{aligned} \tag{A78}$$

We used that $\tilde{R}_{t+1} = 0$ for $t < T$.
For $t = \tau + 1$ and $m = T - t = T - \tau - 1$ we have

$$\kappa(T - \tau - 1, \tau) \; = \; 0 \, , \tag{A79}$$

which characterizes an optimal reward redistribution.

□

Thus, an SDP with an optimal reward redistribution has a expected future rewards that are zero. Equation $\kappa(T - t - 1, t) = 0$ means that the new SDP $\mathcal{P}$ has no delayed rewards as shown in next corollary.

Corollary A1. *An SDP with an optimal reward redistribution fulfills for* $0 \leqslant \tau \leqslant T - t - 1$

$$\mathrm{E}_{\pi} \left[R_{t+1+\tau} \mid s_{t-1}, a_{t-1} \right] \; = \; 0 \, . \tag{A80}$$

The SDP has no delayed rewards since no state-action pair can increase or decrease the expectation of a future reward.

Proof. For $\tau = 0$ we use $\kappa(m, t-1) = 0$ from Theorem A4 with $m = 0$:

$$\mathrm{E}_{\pi} \left[R_{t+1} \mid s_{t-1}, a_{t-1} \right] \; = \; \kappa(0, t-1) \; = \; 0 \, . \tag{A81}$$

For $\tau > 0$, we also use $\kappa(m, t-1) = 0$ from Theorem A4:

$$\begin{aligned}
\mathrm{E}_{\pi} \left[R_{t+1+\tau} \mid s_{t-1}, a_{t-1} \right] &= \mathrm{E}_{\pi} \left[\sum_{k=0}^{\tau} R_{t+1+k} \; - \; \sum_{k=0}^{\tau-1} R_{t+1+k} \mid s_{t-1}, a_{t-1} \right] \\
&= \mathrm{E}_{\pi} \left[\sum_{k=0}^{\tau} R_{t+1+k} \mid s_{t-1}, a_{t-1} \right] \; - \; \mathrm{E}_{\pi} \left[\sum_{k=0}^{\tau-1} R_{t+1+k} \mid s_{t-1}, a_{t-1} \right] \\
&= \kappa(\tau, t-1) \; - \; \kappa(\tau - 1, t-1) \; = \; 0 \; - \; 0 \; = \; 0 \, .
\end{aligned} \tag{A82}$$

□

A related approach is to ensure zero return by reward shaping if the exact value function is known [114].

The next theorem states the major advantage of an optimal reward redistribution: $\tilde{q}^\pi(s_t, a_t)$ can be estimated with an offset that depends only on s_t by estimating the expected immediate redistributed reward. Thus, Q-value estimation becomes trivial and the the advantage function of the MDP $\tilde{\mathcal{P}}$ can be readily computed.

Theorem A5. *If the reward redistribution is optimal, then the Q-values of the SDP $\mathcal{P}$ are given by*

$$\begin{aligned} q^\pi(s_t, a_t) \;&=\; r(s_t, a_t) \;=\; \tilde{q}^\pi(s_t, a_t) \;-\; \mathrm{E}_{s_{t-1}, a_{t-1}} \left[\tilde{q}^\pi(s_{t-1}, a_{t-1}) \mid s_t \right] \\ &=\; \tilde{q}^\pi(s_t, a_t) \;-\; \psi^\pi(s_t) \;. \end{aligned} \tag{A83}$$

The SDP $\mathcal{P}$ and the original MDP $\tilde{\mathcal{P}}$ have the same advantage function. Using a behavior policy $\breve{\pi}$ the expected immediate reward is

$$\mathrm{E}_{\breve{\pi}} \left[R_{t+1} \mid s_t, a_t \right] \;=\; \tilde{q}^\pi(s_t, a_t) \;-\; \psi^{\pi, \breve{\pi}}(s_t) \;. \tag{A84}$$

Proof. The expected reward $r(s_t, a_t)$ is computed for $0 \leqslant t \leqslant T$, where s_{-1}, a_{-1} are states and actions, which are introduced for formal reasons at the beginning of an episode. The expected reward $r(s_t, a_t)$ is with $\tilde{q}^\pi(s_{-1}, a_{-1}) = 0$:

$$\begin{aligned} r(s_t, a_t) \;&=\; \mathrm{E}_{r_{t+1}} \left[R_{t+1} \mid s_t, a_t \right] \;=\; \mathrm{E}_{s_{t-1}, a_{t-1}} \left[\tilde{q}^\pi(s_t, a_t) \;-\; \tilde{q}^\pi(s_{t-1}, a_{t-1}) \mid s_t, a_t \right] \\ &=\; \tilde{q}^\pi(s_t, a_t) \;-\; \mathrm{E}_{s_{t-1}, a_{t-1}} \left[\tilde{q}^\pi(s_{t-1}, a_{t-1}) \mid s_t, a_t \right] \;. \end{aligned} \tag{A85}$$

The expectations $\mathrm{E}_\pi \left[. \mid s_t, a_t \right]$ like $\mathrm{E}_\pi \left[\tilde{R}_{T+1} \mid s_t, a_t \right]$ are expectations over all episodes starting in (s_t, a_t) and ending in some (s_T, a_T).

The Q-values for the SDP $\mathcal{P}$ are defined for $0 \leqslant t \leqslant T$ as:

$$\begin{aligned} q^\pi(s_t, a_t) \;&=\; \mathrm{E}_\pi \left[\sum_{\tau=0}^{T-t} R_{t+1+\tau} \mid s_t, a_t \right] \\ &=\; \mathrm{E}_\pi \left[\tilde{q}^\pi(s_T, a_T) \;-\; \tilde{q}^\pi(s_{t-1}, a_{t-1}) \mid s_t, a_t \right] \\ &=\; \mathrm{E}_\pi \left[\tilde{q}^\pi(s_T, a_T) \mid s_t, a_t \right] \;-\; \mathrm{E}_\pi \left[\tilde{q}^\pi(s_{t-1}, a_{t-1}) \mid s_t, a_t \right] \\ &=\; \tilde{q}^\pi(s_t, a_t) \;-\; \mathrm{E}_{s_{t-1}, a_{t-1}} \left[\tilde{q}^\pi(s_{t-1}, a_{t-1}) \mid s_t, a_t \right] \\ &=\; r(s_t, a_t) \;. \end{aligned} \tag{A86}$$

The second equality uses

$$\begin{aligned} \sum_{\tau=0}^{T-t} R_{t+1+\tau} \;&=\; \sum_{\tau=0}^{T-t} \tilde{q}^\pi(s_{t+\tau}, a_{t+\tau}) \;-\; \tilde{q}^\pi(s_{t+\tau-1}, a_{t+\tau-1}) \\ &=\; \tilde{q}^\pi(s_T, a_T) \;-\; \tilde{q}^\pi(s_{t-1}, a_{t-1}) \;. \end{aligned} \tag{A87}$$

The posterior $p(s_{t-1}, a_{t-1} \mid s_t, a_t)$ is

$$\begin{aligned} p(s_{t-1}, a_{t-1} \mid s_t, a_t) \;&=\; \frac{p(s_t, a_t \mid s_{t-1}, a_{t-1}) \, p(s_{t-1}, a_{t-1})}{p(s_t, a_t)} \\ &=\; \frac{p(s_t \mid s_{t-1}, a_{t-1}) \, p(s_{t-1}, a_{t-1})}{p(s_t)} \;=\; p(s_{t-1}, a_{t-1} \mid s_t) \;, \end{aligned} \tag{A88}$$

where we used $p(s_t, a_t \mid s_{t-1}, a_{t-1}) = \pi(a_t \mid s_t) p(s_t \mid s_{t-1}, a_{t-1})$ and $p(s_t, a_t) = \pi(a_t \mid s_t) p(s_t)$. The posterior does no longer contain a_t. We can express the mean of previous Q-values by the posterior $p(s_{t-1}, a_{t-1} \mid s_t, a_t)$:

$$\begin{aligned} \mathrm{E}_{s_{t-1}, a_{t-1}} \left[\tilde{q}^\pi(s_{t-1}, a_{t-1}) \mid s_t, a_t \right] \;&=\; \sum_{s_{t-1}, a_{t-1}} p(s_{t-1}, a_{t-1} \mid s_t, a_t) \, \tilde{q}^\pi(s_{t-1}, a_{t-1}) \\ =\; \sum_{s_{t-1}, a_{t-1}} p(s_{t-1}, a_{t-1} \mid s_t) \, \tilde{q}^\pi(s_{t-1}, a_{t-1}) \;&=\; \mathrm{E}_{s_{t-1}, a_{t-1}} \left[\tilde{q}^\pi(s_{t-1}, a_{t-1}) \mid s_t \right] \;=\; \psi^\pi(s_t) \;, \end{aligned} \tag{A89}$$

with

$$\psi^\pi(s_t) \;=\; \mathrm{E}_{s_{t-1}, a_{t-1}} \left[\tilde{q}^\pi(s_{t-1}, a_{t-1}) \mid s_t \right] \;. \tag{A90}$$

The SDP $\mathcal{P}$ and the MDP $\tilde{\mathcal{P}}$ have the same advantage function, since the value functions are the expected Q-values across the actions and follow the equation $v^\pi(s_t) = \tilde{v}^\pi(s_t) + \psi^\pi(s_t)$. Therefore $\psi^\pi(s_t)$ cancels in the advantage function of the SDP $\mathcal{P}$.

Using a behavior policy $\breve{\pi}$ the expected immediate reward is

$$\mathrm{E}_{\breve{\pi}}\left[R_{t+1} \mid s_t, a_t\right] = \mathrm{E}_{r_{t+1},\breve{\pi}}\left[R_{t+1} \mid s_t, a_t\right] = \mathrm{E}_{s_{t-1},a_{t-1},\breve{\pi}}\left[\tilde{q}^\pi(s_t, a_t) - \tilde{q}^\pi(s_{t-1}, a_{t-1}) \mid s_t, a_t\right] \tag{A91}$$
$$= \tilde{q}^\pi(s_t, a_t) - \mathrm{E}_{s_{t-1},a_{t-1},\breve{\pi}}\left[\tilde{q}^\pi(s_{t-1}, a_{t-1}) \mid s_t, a_t\right] .$$

The posterior $p_{\breve{\pi}}(s_{t-1}, a_{t-1} \mid s_t, a_t)$ is

$$p_{\breve{\pi}}(s_{t-1}, a_{t-1} \mid s_t, a_t) = \frac{p_{\breve{\pi}}(s_t, a_t \mid s_{t-1}, a_{t-1})\, p_{\breve{\pi}}(s_{t-1}, a_{t-1})}{p_{\breve{\pi}}(s_t, a_t)} \tag{A92}$$
$$= \frac{p(s_t \mid s_{t-1}, a_{t-1})\, p_{\breve{\pi}}(s_{t-1}, a_{t-1})}{p_{\breve{\pi}}(s_t)} = p_{\breve{\pi}}(s_{t-1}, a_{t-1} \mid s_t) ,$$

where we used $p_{\breve{\pi}}(s_t, a_t \mid s_{t-1}, a_{t-1}) = \breve{\pi}(a_t \mid s_t) p(s_t \mid s_{t-1}, a_{t-1})$ and $p_{\breve{\pi}}(s_t, a_t) = \breve{\pi}(a_t \mid s_t) p_{\breve{\pi}}(s_t)$. The posterior does no longer contain a_t. We can express the mean of previous Q-values by the posterior $p_{\breve{\pi}}(s_{t-1}, a_{t-1} \mid s_t, a_t)$:

$$\mathrm{E}_{s_{t-1},a_{t-1},\breve{\pi}}\left[\tilde{q}^\pi(s_{t-1}, a_{t-1}) \mid s_t, a_t\right] = \sum_{s_{t-1},a_{t-1}} p_{\breve{\pi}}(s_{t-1}, a_{t-1} \mid s_t, a_t)\, \tilde{q}^\pi(s_{t-1}, a_{t-1}) \tag{A93}$$
$$= \sum_{s_{t-1},a_{t-1}} p_{\breve{\pi}}(s_{t-1}, a_{t-1} \mid s_t)\, \tilde{q}^\pi(s_{t-1}, a_{t-1}) = \mathrm{E}_{s_{t-1},a_{t-1},\breve{\pi}}\left[\tilde{q}^\pi(s_{t-1}, a_{t-1}) \mid s_t\right] = \psi^{\pi,\breve{\pi}}(s_t) ,$$

with

$$\psi^{\pi,\breve{\pi}}(s_t) = \mathrm{E}_{s_{t-1},a_{t-1},\breve{\pi}}\left[\tilde{q}^\pi(s_{t-1}, a_{t-1}) \mid s_t\right] . \tag{A94}$$

Therefore we have

$$\mathrm{E}_{\breve{\pi}}\left[R_{t+1} \mid s_t, a_t\right] = \tilde{q}^\pi(s_t, a_t) - \psi^{\pi,\breve{\pi}}(s_t) . \tag{A95}$$

□

A2.7 Novel Learning Algorithms based on Reward Redistributions

We assume $\gamma = 1$ and a finite horizon or absorbing state original MDP $\tilde{\mathcal{P}}$ with delayed reward. According to Theorem A5, $\tilde{q}^\pi(s_t, a_t)$ can be estimated with an offset that depends only on s_t by estimating the expected immediate redistributed reward. Thus, Q-value estimation becomes trivial and the the advantage function of the MDP $\tilde{\mathcal{P}}$ can be readily computed. All reinforcement learning methods like policy gradients that use $\arg\max_{a_t} \tilde{q}^\pi(s_t, a_t)$ or the advantage function $\tilde{q}^\pi(s_t, a_t) - \mathrm{E}_{a_t} \tilde{q}^\pi(s_t, a_t)$ of the original MDP $\tilde{\mathcal{P}}$ can be used. These methods either rely on Theorem A5 and either estimate $q^\pi(s_t, a_t)$ according to Eq. (A83) or the expected immediate reward according to Eq. (A84). Both approaches estimate $\tilde{q}^\pi(s_t, a_t)$ with an offset that depends only on s_t (either $\psi^\pi(s_t)$ or $\psi^{\pi,\breve{\pi}}(s_t)$). Behavior policies like "greedy in the limit with infinite exploration" (GLIE) or "restricted rank-based randomized" (RRR) allow to prove convergence of SARSA [118]. These policies can be used with reward redistribution. GLIE policies can be realized by a softmax with exploration coefficient on the Q-values, therefore $\psi^\pi(s_t)$ or $\psi^{\pi,\breve{\pi}}(s_t)$ cancels. RRR policies select actions probabilistically according to the ranks of their Q-values, where the greedy action has highest probability. Therefore $\psi(s_t)$ or $\psi^{\pi,\breve{\pi}}(s_t)$ is not required. For function approximation, convergence of the Q-value estimation together with reward redistribution and GLIE or RRR policies can under standard assumptions be proven by the stochastic approximation theory for two time-scale update rules [17, 64]. Proofs for convergence to an optimal policy are in general difficult, since locally stable attractors may not correspond to optimal policies.
Reward redistribution can be used for

- (A) Q-value estimation,
- (B) policy gradients, and
- (C) Q-learning.